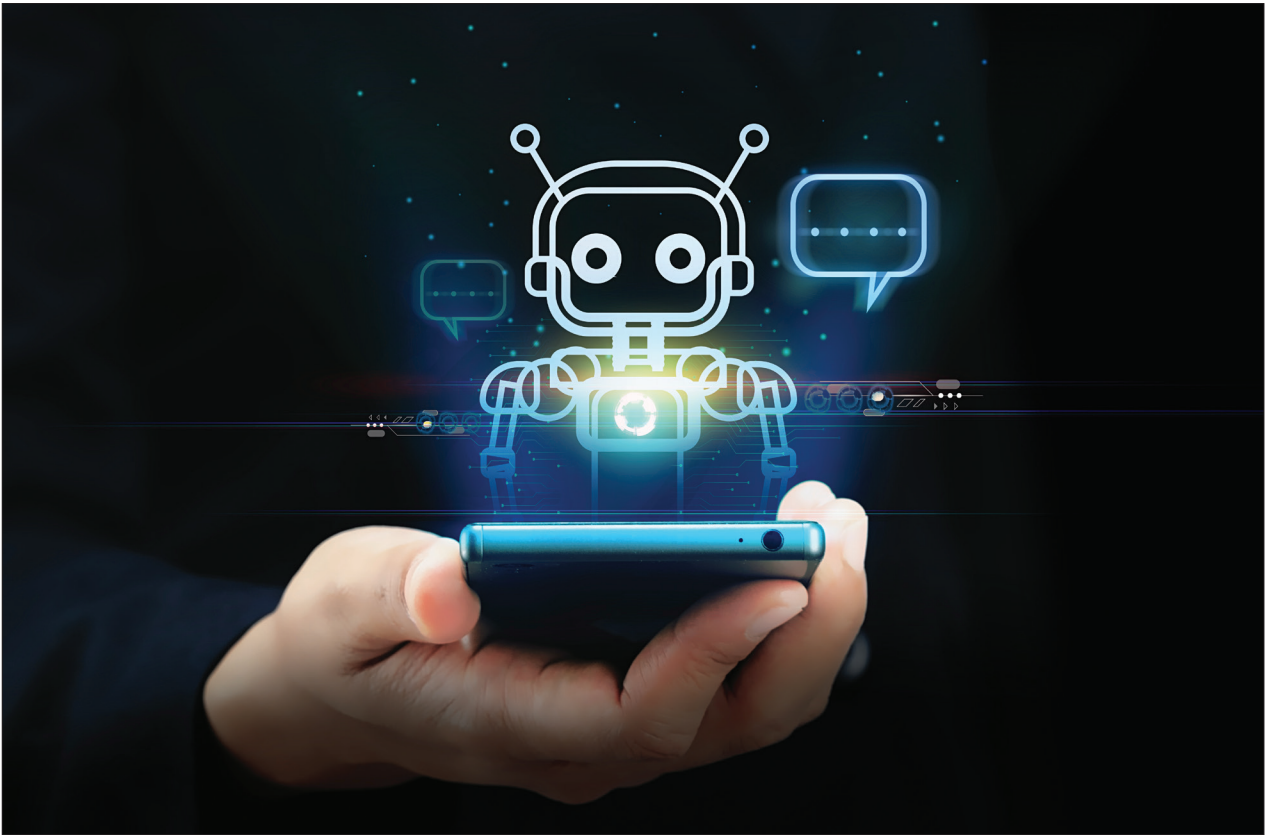




Edge AI

Making AI open for all

Edge AI has become a transformative force in technology, bringing artificial intelligence closer to the devices and environments where data is generated. In 2024, its advancements have enabled significant strides in on-the-go devices, low-latency applications, and security-focused innovations. This chapter explores these developments in depth.

On-the-Go Devices

Smart Glasses with Edge AI

In 2024, smart glasses have evolved significantly, with edge AI enhancing their capabilities. For instance, Vuzix's latest models integrated real-time facial recognition and augmented reality (AR) overlays. By processing visual information locally, these devices eliminated reliance on cloud servers, ensuring faster responses and enhanced privacy. These glasses were particularly

useful in professional settings such as healthcare and field services, where rapid, secure data interpretation was critical. According to industry reports, the global AR smart glasses market is projected to reach \$8.2 billion by 2025, driven by innovations like edge AI.

Edge AI for Retail

Retail environments saw remarkable innovation with edge AI's adoption in inventory management. Walmart, for example, deployed AI cameras and sensors in their stores to analyze shelf stock in real-time. This localized data processing ensured shelves were replenished promptly without sending large datasets to centralized servers. Such systems improved operational efficiency while minimizing bandwidth use and cloud dependency. A Deloitte study highlighted that retailers using edge AI experienced a 25% increase in inventory accuracy and a 20% reduction in restocking delays in 2024 alone.